Research Proposal

Towards Global Workspace Theoretic Models

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Abstract

Despite remarkable advances in scale and capability, Transformer-based models remain brittle in long-context reasoning and systematic generalisation. As sequence length increases, task-relevant information can become inaccessible, suggesting limitations in how current architectures coordinate and preserve representations across layers and time. We observe that standard Transformers lack an explicit mechanism for selectively consolidating task-relevant information and broadcasting it globally across the architecture; instead, all communication is funnelled through a dense, unstructured residual stream. Global Workspace Theory (GWT), a well-established cognitive framework, proposes that higher-order cognition arises precisely from such a mechanism: distributed specialist sub-processes compete for access to a capacity-limited shared workspace, whose contents are then broadcast system-wide. In this project, we propose to operationalise the core components of GWT, namely selective attention and structured global broadcasting, as explicit architectural mechanisms within Transformer-based models. We aim to evaluate whether this principled separation of local processing and global integration improves performance on long-context reasoning, compositional generalisation, and few-shot learning tasks, and to conduct mechanistic analyses investigating how broadcasting alters internal representational dynamics.

Central Research Question: can (structured) global broadcasting mechanisms improve information persistence and reasoning across long contexts in transformer-based models.

1 Background and Motivation

Recent advances in large-scale Transformer-based models [Vaswani et al., 2017] have led to remarkable improvements in language modelling, recall, and pattern recognition [Brown et al., 2020, Touvron et al., 2023]. Large Language Models (LLMs) demonstrate strong performance on factual recall and even certain forms of compositional reasoning. However, despite scaling improvements, persistent challenges remain in settings requiring robust few-shot generalisation, long-range reasoning, and structured information integration across extended contexts [Bai et al., 2025, Li et al., 2024]. Performance often degrades with increasing sequence length, and information introduced in the middle of such a context can become effectively inaccessible during downstream reasoning [Levy et al., 2024, Liu et al., 2023].

These limitations suggest that current architectures may lack mechanisms for maintaining and coordinating task-relevant information over time. In standard Transformer architectures, attention provides a flexible mechanism for pairwise token interaction, yet it does not impose explicit structure on how information is globally consolidated or selectively broadcast across layers and time steps [Elhage et al., 2021, Olsson et al., 2022]. As a result, models rely heavily on implicit coordination via attention weights and residual pathways, which may be insufficient for structured, persistent reasoning [Elhage et al., 2021, Wang et al., 2022].

In contrast, cognitive frameworks, such as Global Workspace Theory (GWT) [Baars, 1988, 2005], propose that high-level cognition emerges from the selective broadcasting of information across distributed specialist processes. According to GWT, many parallel subsystems operate semi-independently, but task-relevant representations are periodically amplified and globally broadcast, enabling coordinated integration and sustained reasoning. Central to this framework are two mechanisms: (i) selective attention, which determines which representations gain access to the workspace, and (ii) global broadcasting, which disseminates selected information across the system.

While Transformer architectures include components loosely reminiscent of these mechanisms, such as multi-head attention and residual connections, there is currently no explicit architectural analogue of a global workspace with controlled broadcasting dynamics. Existing implementations remain largely homogeneous, with no clear separation between local processing and globally shared representational states. Consequently, it remains unclear whether introducing structured global broadcasting mechanisms could improve information persistence, coordination across layers, and reasoning over long contexts.

This gap motivates the present work: to investigate whether principled architectural mechanisms inspired by selective attention and global broadcasting can enhance long-context reasoning and structured generalisation in Transformer-based models.

1.1 Related Works

A fuller review of related work is provided in Appendix A. Here we highlight the most directly relevant lines of work.

The architecture most closely aligned with our goals is the Shared Workspace of Goyal et al. [2022], in which specialist modules (sequence positions in the case of Transformers) write to a capacity-limited workspace via competitive attention and read consolidated information back. This work demonstrated improved systematic generalisation and coordination across modular architectures including Transformers, but was evaluated at relatively small scale (few layers, small hidden dimensions, short sequences) and not on long-context language tasks where the benefits of such a mechanism may be most consequential. Furthermore, their modified Transformer introduces sparse contributions at the sequence level but includes contributions across all layers and heads. This gap further motivates both our problem setting and our choice of an attention-based selection mechanism within a Transformer-inspired architecture.

Other GWT-inspired proposals include the Global Latent Workspace of VanRullen and Kanai

[2021], which remains largely conceptual, and the embodied GWT agent of Dossa et al. [2024], which targets multimodal navigation rather than language reasoning. Most recently, Chateau-Laurent and VanRullen [2025] implement a GWT-inspired architecture in which a controller selectively routes information between specialised modules through a shared workspace via a gating mechanism. Their model outperforms both LSTMs and Transformers on a sequential addition task requiring length generalisation, providing initial evidence that workspace-mediated broadcasting can improve systematic reasoning. However, the evaluation remains limited to a simple arithmetic task with hand-designed or minimally learned modules, and does not address language or long-context settings. The theoretical grounding for all of these traces to the consciousness prior of Bengio [2017] and the broader inductive bias programme of Goyal and Bengio [2022].

Standard Transformer architectures [Vaswani et al., 2017] develop specialised sub-components - induction heads [Olsson et al., 2022], factual recall circuits [Meng et al., 2022] - but funnel all communication through a single residual stream [Elhage et al., 2021], which lacks both sparse selection and a distinct shared workspace. Several extensions partially address this: mixture-of-experts (MoE) models [Fedus et al., 2022, Shazeer et al., 2017] add routing sparsity but not a separate workspace; memory and thinking tokens [Burtsev et al., 2021, Goyal et al., 2024, Hao et al., 2025] gesture towards auxiliary processing but remain embedded within the residual stream; dense cross-layer connections [Heddes et al., 2025, Pagliardini et al., 2024, Xiao et al., 2025] improve information flow but do so with dynamic yet dense connections rather than selective consolidation to a global workspace; and finally, the Perceiver [Jaegle et al., 2021] introduces a latent bottleneck but lacks competitive multi-module access.

Evaluated against the three core characteristics of a global workspace - a shared representational state, sparse competitive contributions, and bidirectional access - no existing architecture fully delivers. This project proposes to close this gap within Transformer-based architectures for language, and to rigorously evaluate whether such structure improves long-context reasoning and generalisation.

2 Research Objectives

- **RQ1: Formalisation.** How can selective attention and global broadcasting, as described in Global Workspace Theory, be operationalised as explicit architectural mechanisms within Transformer-based models?
- **RQ2: Architectural efficacy.** Do structured global broadcasting mechanisms improve information persistence and long-context reasoning compared to standard attention-based architectures?
- **RQ3: Mechanistic understanding.** Under what conditions do broadcasting mechanisms enhance coordination across layers and time, and how do these mechanisms alter representational dynamics within the model?
- **RQ4: Generalisation.** Can structured broadcasting improve few-shot learning and task

generalisation beyond improvements attributable to scale alone?

3 Proposed Methodology

The central hypothesis underlying this project is that an explicit architectural separation between *local processing* and *global integration* will improve structured reasoning by enhancing the persistence and coordination of task-relevant information across layers and time. Standard Transformer architectures rely on homogeneous self-attention and residual pathways for coordination, but do not impose explicit structure on how information is globally consolidated or selectively broadcast. This project investigates whether introducing principled global workspace mechanisms can improve long-context reasoning and generalisation.

The methodology proceeds in three interconnected stages: (i) architectural formalisation, (ii) empirical evaluation, and (iii) mechanistic analysis.

3.1 Architectural Formalisation of a Global Workspace

The first stage focuses on translating the conceptual components of Global Workspace Theory, selective attention and global broadcasting, into explicit architectural mechanisms based around transformers.

Workspace State Representation

A structured global workspace will be introduced as a shared representational state accessible across layers and time. The Shared Workspace (SW) proposed by Goyal et al. [2022] is a natural initial candidate, having already demonstrated consistent improvements in generalisation capabilities while remaining highly compatible with the Transformer architecture. Extending this mechanism to be selectively sparse across layers and heads would further strengthen its resemblance to GWT, broadening its use of "specialist" representations from sequence positions to model components.

We will then separate contributions to and from the workspace from the standard residual pathway of the Transformer, creating a distinction between local, distributed processing (standard self-attention) and global integration and broadcast dynamics (shared workspace). This separation enables investigation into whether structured global consolidation improves information persistence over long contexts.

Selective Access Mechanisms

Selective access to the workspace will be implemented through sparse and competitive selection mechanisms. Rather than allowing all components to contribute equally, candidate representations and outputs will compete for entry into the global state. Existing methods include the use of cross-attention across components: the SW can be viewed as employing such a mechanism

between individual layers and the shared workspace; alternatively, ElNokrashy et al. [2024] propose attending directly to all past layer representations. Such methods will be extended such that a single head could selectively contribute to the state by being paired with complementary mechanisms. For example, learned gating functions, popular in MoE models for sparse routing, could be used to determine which layers and heads should contribute to the workspace depending on context.

Structured Global Broadcasting

Once updated, the workspace state will be broadcast back to local processing streams. Broadcasting may occur across all layers and time-steps, or exclude certain parts dedicated to initial, dense processing. The key idea within GWT is that such a state is shared by all contributing components; accordingly, any component with the potential to write to the state should also receive it as input. We will investigate what proportion of the model should be subject to such mechanisms, and whether the workspace state should replace inputs to contributing components entirely or be combined selectively with the residual stream via gating mechanisms. The former would then have the potential to be parallelised across components, between updates to the workspace, potentially improving inference speeds with the use of modern hardware. On the other hand, the latter may further encourage greater collaboration between sub-components. Furthermore, we will compare these approaches with that of the SW, which uses cross-attention directly between layers and the workspace. Finally, bi-directional access may then be achieved through recurrence of sub-components, for example. The design enables investigation into whether explicit global consolidation enhances reasoning robustness and reduces representational interference.

3.2 Expanding Attention Beyond Block-Level Self-Attention

To support structured coordination, the project will explore extensions of attention mechanisms beyond the standard intra-layer self-attention common in Transformer architectures. Initial methods for such extensions have already been introduced in Section 3.1 for both inputs to and outputs from the shared workspace. However, these methods have so far been limited to interactions with the shared workspace alone. To confirm the necessity of a constrained, shared state for encouraging cross-component collaboration and thereby improving generalisation, we will ablate such attention-based mechanisms without the shared workspace. Specifically, we will implement, in parallel with our target architectures, a matching mechanism that bypasses the workspace entirely and facilitates sparse connections between individual model components. Prior work has demonstrated success in this regard through weighted skip connections [Heddes et al., 2025, Pagliardini et al., 2024, Xiao et al., 2025]. We will use these methods as baselines and further extend them with our own cross-component attention mechanisms, mirroring any methods used to interact with the shared workspace.

3.3 Empirical Evaluation

Architectural variants will be evaluated on tasks requiring long-context reasoning, few-shot generalisation, and structured information integration across extended sequences.

For long-context reasoning and cross-document integration, we will employ RULER [Hsieh et al., 2024], BABILong [Kuratov et al., 2024], and Loong [Wang et al., 2024], which collectively test retrieval, multi-hop tracing, fact chaining, and multi-document reasoning at context lengths scaling from thousands to millions of tokens. Together, these allow us to evaluate whether structured broadcasting mechanisms improve information persistence as context length and reasoning complexity increase. For few-shot and compositional generalisation, we will use COGS [Kim and Linzen, 2020], and CFQ [Keysers et al., 2020], established benchmarks on which standard transformers are known to struggle with length and structural generalisation splits. These are especially relevant, as the global workspace hypothesis predicts that structured cross-component integration should yield more systematic recombination of learned primitives. For broader structured integration, we will draw on LongBench v2 [Bai et al., 2025], which spans six task categories, including multi-document QA, long in-context learning, and code repository understanding with contexts up to 2M words.

Comparisons will be made against baseline Transformer architectures of comparable scale to isolate the effects of structured broadcasting mechanisms. In the event of a complex, multi-component system, each contributing method will be ablated alongside the closest corresponding component found in current Transformer architectures. Evaluation will focus not only on performance metrics, but also on robustness to model scale, increasing context length, and susceptibility to interference from distractor information.

3.4 Mechanistic and Representational Analysis

Beyond task-level performance, the project will investigate how structured broadcasting mechanisms alter internal model dynamics, focusing on information persistence across layers, attention routing behaviour, and representational interference. While exact methods will depend on the final architectures, we draw on established and recent techniques from mechanistic interpretability [Rai et al., 2025].

To measure information content and persistence, we will train linear probing classifiers [Belinkov, 2021] on intermediate representations to track whether task-relevant features are retained over greater depth in workspace-augmented models, complemented by the tuned lens [Belrose et al., 2025] to compare prediction trajectories across architectures. We will additionally track attention entropy across layers, testing whether workspace augmentation attenuates the progressive sparsification of attention scores observed in standard Transformers [Zhang et al., 2023], which we hypothesise reflects increasing representational interference in the residual stream.

To distinguish whether workspace broadcasting induces genuine collaborative computation or merely redundant duplication, we will apply partial information decomposition [Mediano et al., 2021, Williams and Beer, 2010] to attention head activations. Recent work has shown that LLMs spontaneously develop a synergistic core in their middle layers with a redundant periphery in

early and late layers [Luppi et al., 2024, Urbina-Rodriguez et al., 2026]; we will test whether workspace augmentation reshapes this informational architecture, for example by extending synergistic processing across a greater proportion of layers.

To establish causal rather than correlational claims, we will employ activation patching [Heimersheim and Nanda, 2024, Meng et al., 2022], replacing workspace states with corrupted or baseline-derived activations to determine whether global broadcast causally mediates task performance. Where feasible, automated circuit discovery [Conmy et al., 2023] will identify whether workspace-augmented models develop qualitatively distinct computational sub-graphs, and centred kernel alignment [Kornblith et al., 2019] will quantify changes in representational geometry across variants.

The objective is to determine whether explicit workspace mechanisms induce qualitatively different coordination patterns compared to standard Transformer architectures, clarifying the computational role of structured global broadcasting.

4 Expected Contributions and Impact

This project is expected to make contributions at three complementary levels: (1) architectural innovation, (2) mechanistic understanding, and (3) broader conceptual integration between cognitive theory and machine learning.

4.1 Architectural Contributions

The primary technical contribution will be the design and evaluation of Transformer-based architectures incorporating explicit structured global broadcasting mechanisms. By introducing a principled separation between local processing and global integration, the project aims to establish whether such architectural structure improves information persistence, coordination, and long-context reasoning.

If successful, this work will provide:

- A formal architectural framework for implementing global workspace-inspired mechanisms within Transformer models,
- Empirical evidence regarding the conditions under which structured broadcasting improves reasoning and generalisation,
- Practical design insights for scalable long-context and few-shot learning systems.

These contributions would extend existing approaches to memory augmentation and attention modification by introducing explicit global coordination dynamics rather than relying solely on homogeneous self-attention.

4.2 Mechanistic Understanding of Coordination in Transformers

Beyond performance improvements, a central contribution of this work lies in mechanistic analysis. By studying how explicit broadcasting mechanisms alter internal model dynamics, the project aims to clarify the role of global integration in deep neural networks.

This includes identifying:

- Whether structured workspace states enhance the persistence of task-relevant information,
- How selective competition alters representational dynamics,
- Whether explicit separation of local and global pathways reduces interference across long contexts.

Such analysis may contribute to a deeper understanding of coordination and information flow in Transformer architectures, moving beyond empirical benchmarking toward principled architectural interpretation.

4.3 Bridging Cognitive Theory and Machine Learning

A further contribution lies in the systematic operationalisation of concepts drawn from Global Workspace Theory within modern machine learning systems. Rather than treating cognitive frameworks solely as metaphorical inspiration, this project seeks to translate key components of such a framework into explicit computational mechanisms.

By evaluating these mechanisms under controlled experimental settings, the work may clarify which aspects of cognitive workspace theory have computational relevance for scalable neural architectures, and which do not. This provides a structured pathway for integrating cognitive theory into machine learning research in a manner that is empirically grounded and technically rigorous.

4.4 Broader Impact

Improving structured reasoning and long-context information coordination has implications for a range of applications, including scientific text analysis, multi-step planning, and decision-making systems operating over extended inputs. Architectures capable of more robust information persistence may also contribute to improved reliability in continual learning settings, thereby extending the lifetime of a machine learning model, which remains otherwise expensive to train or fine-tune.

More broadly, this research may contribute to the development of machine learning systems that exhibit continuous, and improved generalisation under limited supervision, potentially reducing dependence on scale alone as the primary driver of performance gains.

5 Feasibility and Timeline

The project is structured over three to four years, progressing from conceptual grounding and architectural development toward empirical validation and consolidation.

Year 1: Foundations and Initial Prototyping

- Comprehensive literature review across long-context Transformer architectures, memory and continual learning mechanisms, and Global Workspace Theory.
- Formalisation of workspace-inspired architectural principles.
- Implementation of initial prototype models incorporating explicit global workspace components.
- Preliminary experiments evaluating information persistence and long-context behaviour.

The primary objective of the first year is to establish a clear computational formulation of structured broadcasting mechanisms and validate feasibility through initial, controlled experiments.

Year 2: Architectural Refinement and Systematic Evaluation

- Iterative refinement of workspace and selective access mechanisms.
- Systematic ablation studies comparing architectural variants.
- Evaluation on long-context reasoning and few-shot generalisation tasks.

This stage focuses on establishing empirical evidence for the benefits and limitations of structured global broadcasting mechanisms.

Year 3: Mechanistic Analysis and Theoretical Integration

- In-depth analysis of representational dynamics and coordination patterns within trained models.
- Investigation of conditions under which broadcasting improves generalisation.
- Theoretical consolidation linking architectural findings to broader principles of information integration.
- Preparation of final publications and integration of results into a coherent thesis narrative.

The emphasis in this phase shifts from architectural performance to mechanistic understanding and conceptual synthesis.

Year 4: Consolidation and Thesis Completion

- Final refinements and potential extension studies with aim to close any gaps in the overall narrative.
- Completion and submission of the doctoral thesis.

6 Fit with UCL and Supervision

This project is well aligned with UCL's core research priorities. The UCL Centre for Artificial Intelligence takes an explicitly interdisciplinary approach, drawing on "computer science, statistical learning theory, probabilistic modelling, engineering, cognitive neuroscience and the social sciences" [UCL Centre for Artificial Intelligence, 2025]. It further identifies Interpretation as one of its six fundamental research pillars: a commitment reflected in active doctoral research on mechanistic interpretability within the affiliated UKRI CDT in Foundational AI. The Gatsby Computational Neuroscience Unit, founded to study the mathematical basis of intelligent behaviour in both natural and artificial systems [Gatsby Computational Neuroscience Unit, 2025], anchors the neuroscience–machine learning intersection at UCL, with ongoing research spanning learning dynamics in deep networks and the theoretical analysis of attention-based architectures. UCL's broader institutional vision further reinforces this alignment: its "AI for People and Planet" programme identifies neuroscience-inspired methods as central to foundational AI research [UCL AI for People and Planet, 2025], while the recently established Gen AI Hub specifically targets the understanding of reasoning abilities in language models [UCL EPSRC AI Hub in Generative Models, 2025]. Together, these centres provide an environment in which operationalising cognitive theory within modern deep learning architectures, and subjecting the resulting mechanisms to rigorous mechanistic analysis, is both feasible and strategically encouraged.

I have been working as a Research Engineer with Professor Jun Wang, and more closely with Dr Zafeirios Fountas, on neuroscience-inspired ML models for the past three years since my masters thesis in the summer of 2023. Our research has directly addressed several of the core components that this project seeks to integrate, including memory systems (episodic [Benfeghoul et al., 2025a, Fountas et al., 2024, Wang, 2025, Zhou et al., 2025] and compressive [Benfeghoul et al., 2025b]), iterative reasoning [Benfeghoul et al., 2024], continual learning [Hazard et al., 2025, Wang, 2025, Zhou et al., 2025], and hierarchical models [Zakharov et al., 2021, 2022]. This body of work has consistently operated at the intersection of cognitive science and machine learning, grounding neuroscience-inspired architectural principles in rigorous empirical evaluation. Such an approach aligns closely with the interdisciplinary research environment outlined above, and that positions the proposed supervisory team to guide this project from conceptual formulation through to mechanistic analysis.

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A Extended Related Works

Baars [2005] presents consciousness functions as a global information-sharing mechanism in a brain made of many specialised, largely unconscious processors. Conscious contents are those that gain access to a "global workspace", allowing them to be broadcast widely across the brain. This broadcast enables coordination, integration, control, learning, and reportability.

GWT necessitates a few key characteristics:

1. Working memory: a shared, or "global" state which multiple subprocesses of a model can access
2. Attention: contributions to said global state are sparse and selective, directed by a notion of saliency
3. Bidirectional access: subprocesses can both read from and write to the shared state

A.1 GWT in ML

This section situates the proposed research within four interconnected areas: (i) existing architectures evaluated against the core characteristics of a global workspace theoretic model -

namely a shared workspace, sparse selective contributions, and bidirectional (two-way) access - (ii) Global Workspace Theory and its existing connections to machine learning, (iii) memory augmentation and auxiliary token mechanisms, and (iv) mechanistic interpretability of information flow in transformers.

A.1.1 Architectural Landscape: Towards a Global Workspace

A useful lens through which to evaluate existing architectures is the set of properties that Global Workspace Theory identifies as essential for coordinated cognition: (a) a shared workspace distinct from local processing, (b) sparse and competitive selection of what enters it, and (c) bi-directional access whereby sub-components both contribute to and read from the workspace. No dominant architecture currently satisfies all three criteria, with most failing to meet any of them.

Recurrent Architectures Classical RNNs [Jordan, 1986, Rumelhart et al., 1985], including LSTMs [Hochreiter and Schmidhuber, 1997] and GRUs [Cho et al., 2014], maintain a hidden state that is updated and reused at each time step, making them natural candidates for state recycling. However, this state is not shared among independent sub-systems; rather, it is a monolithic representation updated by a single recurrent function. The hidden state thus acts as a *recycled* workspace but not a *shared* one - there is no notion of multiple specialist modules competing for access. Gating mechanisms (forget, input, output gates) provide a form of selective updating, but the selection operates uniformly over the hidden state rather than mediating between independent processing streams.

State Space Models Structured state space models (SSMs), including S4 [Gu et al., 2022] and Mamba [Gu and Dao, 2023], offer linear-time sequence modelling through a compressed recurrent state. Like RNNs, they maintain a state that is carried forward through time, but they lack independent sub-systems whose outputs compete for workspace access. Mamba's selective scan mechanism introduces input-dependent gating of the state transition, allowing the model to selectively propagate or forget information along the sequence. This addresses content-awareness - a key weakness of earlier time-invariant SSMs - but the selection operates within a single processing channel rather than arbitrating between multiple specialist modules. Recurrent SSMs thus share the RNN limitation: they recycle state but do not broker it to and from independent components.

Transformers and the Residual Stream Bottleneck Transformers [Vaswani et al., 2017] present a more nuanced picture. Attention heads and MLP sub-layers have been shown to specialise for distinct functions - induction heads for in-context copying [Olsson et al., 2022], factual recall circuits in MLP layers [Meng et al., 2022], and various task-specific attention patterns [Voita et al., 2019]. In this sense, transformers do develop semi-independent specialist sub-components.

However, all such components communicate exclusively through a single residual stream [Elhage et al., 2021]. As described in the mathematical framework of Elhage et al. [2021], every attention head and MLP layer reads from and writes to this shared linear channel. The residual stream can thus be viewed as an *implicit*, and *unstructured* workspace. Crucially, however, it lacks two of the three characteristics: contributions are not sparse (every layer writes to the full stream at every step), and there is no competitive selection mechanism governing which representations gain global access. Furthermore, the residual stream is under enormous bandwidth pressure: the number of features that need to be represented vastly exceeds the residual stream's dimensionality, forcing the model into superposition [Bricken et al., 2023, Elhage et al., 2022]. This creates representational interference that may degrade structured reasoning over long contexts.

Mixture-of-Experts. MoE architectures [Fedus et al., 2022, Lepikhin et al., 2020, Shazeer et al., 2017] introduce sparsity into the processing pipeline through learned routing mechanisms that activate only a subset of expert modules per token. This addresses the sparseness criteria - not all sub-systems contribute to every input. However, MoE routing operates at the level of individual feed-forward experts within a layer; it does not create a separate shared workspace, and routed outputs are still funnelled back through the same residual stream. The routing is also typically one-directional: experts do not read from a consolidated global state before contributing their output.

Memory and Thinking Tokens. A separate line of work introduces auxiliary tokens that operate outside the primary predictive sequence. Memory tokens [Burtsev et al., 2021] are processed alongside input tokens, and can even serve as a persistent information store across segments [Bulatov et al., 2022]. Thinking or pause tokens [Goyal et al., 2024, Hao et al., 2025, Herel and Mikolov, 2024, Pfau et al., 2024] provide the model with additional computation steps not tied to predicting specific output tokens, enabling a form of internal deliberation. These approaches move towards a workspace that is separate from the main input/output sequence, and thinking tokens in particular encourage the model to consolidate information before committing to a prediction. However, the information in these auxiliary tokens still flows through the same residual stream and attention mechanism as all other tokens - they do not escape the residual stream bottleneck, nor do they introduce sparse, competitive selection over what enters or leaves this auxiliary space.

Skip and Dense Connections. Skip connections [He et al., 2016] and their generalisations - DenseFormer [Pagliardini et al., 2024], Hyper-Connections [Zhu et al., 2025], DeepCrossAttention [Heddes et al., 2025], and MUDDFormer [Xiao et al., 2025] - provide pathways for information to bypass intermediate layers, partially alleviating the residual stream bottleneck by enabling richer cross-layer information flow. These approaches improve depth utilisation and reduce the rank collapse associated with deep residual networks. However, they maintain a degree of interference: connections are typically dense (all layers contribute to all subsequent layers) and, critically, do not contribute globally. There is no mechanism to determine which layer outputs are task-relevant and should be globally broadcast versus which should remain local. The result is improved gradient flow and information mixing, but not the structured, competitive consolidation that characterises a global workspace.

The Perceiver as a Partial Workspace The Perceiver [Jaegle et al., 2021] and Perceiver IO [Jaegle et al., 2022] are architecturally closest to a workspace model. They use a small set of learned latent variables to compress and integrate high-dimensional inputs via cross-attention, then decode back to the desired output space. The latent bottleneck functions as a compressed workspace, and the cross-attention mechanism provides a form of two-way access (encoding into and decoding from the latent space). However, as noted by Goyal et al. [2022], the Perceiver's latent bottleneck iteratively queries positional information rather than mediating between independent specialist modules; there is no notion of multiple sub-systems competing for access to a shared state. The compression is driven by efficiency rather than by the GWT principle of selective amplification and broadcast.

A.1.2 Global Workspace Theory in Cognitive Science and Machine Learning

Global Workspace Theory, originally proposed by Baars [1988] and extended neurally by Dehaene and Naccache [2001], Dehaene et al. [1998], posits that conscious cognition arises from the selective amplification and global broadcasting of information from parallel specialist subsystems to a capacity-limited workspace.

In the machine learning context, Bengio [2017] articulated the "consciousness prior": the hypothesis that high-level representations should be low-dimensional and sparse, framing consciousness as an information bottleneck that selects and broadcasts task-relevant content. This perspective connects to a broader theoretical tradition: Information Bottleneck theory [Tishby et al., 2000] posits that generalisation emerges from an optimal balance between compressing input information and preserving predictive content in latent representations, yielding generalisation bounds that scale with the tightness of such bottlenecks in deep learning models [Kawaguchi et al., 2023]. Recent work has applied this principle directly to Transformers [Oomerjee et al., 2025], improving generalisation on mathematical reasoning tasks. Goyal and Bengio [2022] expanded the consciousness prior into a broader programme connecting System 2 reasoning to inductive biases including modularity, sparse communication, and bottlenecked coordination. Their "Coordination Among Neural Modules Through a Shared Global Workspace" [Goyal et al., 2022] is the most direct implementation: specialist modules (either Transformer-based or recurrent) write to a limited-capacity shared workspace via competitive attention, and read consolidated information back. This work demonstrated that capacity-limited workspaces can improve systematic generalisation across multiple modular architectures.

Juliani et al. [2022] provided a systematic mapping of GWT's functional properties onto deep learning components, arguing that the Perceiver architecture partially instantiates a global workspace. VanRullen and Kanai [2021] proposed a Global Latent Workspace based on unsupervised neural translation between module-specific latent spaces, emphasising cycle-consistent broadcasting as the mechanism for cross-module integration. More recently, Dossa et al. [2024] designed and evaluated a global workspace agent satisfying the GWT "indicator properties" outlined by Butlin et al. [2023], testing it in an embodied audio-visual navigation environment - the most realistic setting in which a GWT agent has been evaluated. Most recently, Chateau-Laurent and VanRullen [2025] implement a GWT-inspired architecture in which a controller

selectively routes information between specialised modules through a shared workspace via a gating mechanism. Their model outperforms both LSTMs and Transformers on a sequential addition task requiring length generalisation, providing initial evidence that workspace-mediated broadcasting can improve systematic reasoning. However, the evaluation remains limited to a simple arithmetic task with hand-designed or minimally learned modules, and does not address language or long-context settings.

While these works provide valuable groundwork, several gaps remain. Goyal et al. [2022] evaluated their workspace on relatively small-scale modular systems (slot-based models, RIMs) rather than at transformer scale with long-context language tasks. VanRullen and Kanai's proposal remains largely conceptual. Dossa et al. [2024] focused on embodied perception rather than language reasoning. The present project aims to address these gaps by: (a) implementing structured broadcasting mechanisms within transformer-scale architectures for language, (b) systematically ablating individual components to isolate their contributions, and (c) performing mechanistic analysis of how these mechanisms alter representational dynamics - moving from conceptual analogy and small-scale demonstrations to empirically grounded architectural principles for long-context reasoning and generalisation.

A.1.3 Mechanistic Interpretability and Information Flow

Work in mechanistic interpretability provides both motivation and analytical tools for this project. Research on induction heads [Olsson et al., 2022], circuit-level analysis [Elhage et al., 2021, Wang et al., 2022], and the superposition hypothesis [Bricken et al., 2023, Elhage et al., 2022] has revealed that transformers develop structured internal mechanisms for copying, retrieval, and composition, all communicated through the residual stream.

The observation that the residual stream functions as an implicit workspace - and that it is under severe bandwidth pressure, forcing representations into superposition - directly motivates the present work. If the residual stream already serves as an unstructured global channel but suffers from interference due to non-selective, dense contributions, then introducing an *explicit* workspace with sparse, competitive, and bi-directional access may reduce this interference while preserving the coordination benefits. The mechanistic analysis component of this project (RQ3) is designed to test this hypothesis directly, using probing classifiers, representational similarity analysis, and attention pattern analysis to compare information persistence and coordination in standard transformers versus architectures with explicit broadcasting mechanisms.

A.1.4 Summary and Positioning

The proposed research sits at the intersection of these areas. Evaluated against the three core characteristics of a global workspace - a shared representational state, sparse competitive contributions, and bidirectional access - no existing architecture fully delivers. RNNs and SSMs recycle state but lack independent sub-systems. Transformers develop specialist components but bottleneck all communication through a single, non-selective residual stream. MoE models add routing sparsity but not a separate workspace. Memory and thinking tokens gesture towards

an auxiliary workspace but remain embedded within the residual stream. Skip connections improve cross-layer flow but within a forward-only, cumulative selection of past outputs rather than a global shared state. The Perceiver comes closest structurally but lacks competitive, multi-module access. This project proposes to close this gap: introducing an explicit, structured global workspace with sparse bidirectional access into transformer-based architectures, and rigorously evaluating whether such structure yields measurable improvements in information persistence, long-context reasoning, and generalisation.

B Preliminary Ideas, Architectures, and Ablations

B.1 Ideas and Directions

An advantage of our current problem formulation is both the reliance on- and interaction between structured sub-systems, and the contributions of said methods to information persistence and reasoning capabilities. This ensures that relevant topics and directions include memory and recall, consolidation and nested learning, reasoning, and other directions that are highly related to our current research directions and objectives.

B.2 Initial Architectures: Global Transformers

The idea behind this initial architecture is to test whether actively persisting and re-using a subset of information from earlier layers at various points within the Transformer architecture can improve performance. The key mechanism behind it is simply to designate a single KV head as a "Global Head" which will be re-used in a later attention block, as illustrated in Fig. 1, with new query heads generated from the residual stream. The intuition behind it is that key information from specialised earlier layers may be lost within the residual stream but still be relevant to later layers. Essentially, we want to remove the constraint of a forward-only model by introducing cross-attention at various checkpoints in the architecture.

Pros:

- Selective contributions to a global state: re-processing states from different layers with new context.
- Intrinsic to the Transformer architecture: relies on existing kernels while also being more parameter and compute-efficient than its Transformer counterpart.
-
- Bi-directional access: recurrence allows contributing layers to get access to this global state after contribution within a single pass of the model, extending the bi-directional access characteristic otherwise only maintained across time-steps.

Cons:

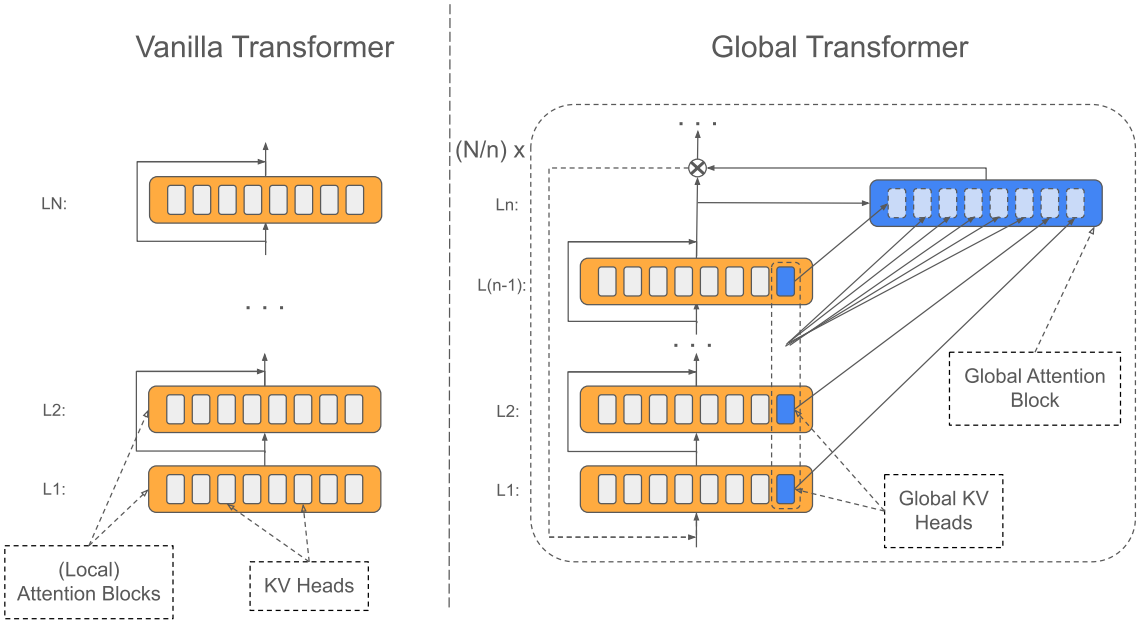


Figure 1: Illustrating the basis of the Global Transformer architecture. Global Attention blocks are added after n local (a.k.a vanilla) attention blocks. The blocks recycle a single KV head from each previous local block and perform cross-attention across them with its own Query heads. Note that, given the same total number of attention blocks, local and global, this architecture means less parameters and compute than the Vanilla Transformer due to the recycling of KV heads. Finally, the use of recurrence can both reduce the number of parameters drastically and better align with GWT by re-using the global state as input (bi-directional access).

- Dependent sub-systems: in the current formulation, the input of every layer (apart from the first) is potentially dependent on the layers that came before it, depending on how much they contributed to the residual stream.

B.2.1 Alternatives to Recurrence: Sequential vs. Parallel

B.2.2 Ablation: Dense Global Transformers

This design should extend the new (and baseline) architectures with the use of dynamic dense connections (see Section A.1.1), in order to investigate whether such connections between sub-components can improve performance. As briefly mentioned in Section 3.2, we will seek to investigate whether escaping the residual in the inputs of the underlying layers further improve the architecture or whether a constrained, global workspace is strictly necessary in order to achieve our desired improvements in model performance.

B.3 Initial Architectures: Memory-based

An alternative, more structured approach would instead mediate component interactions through a separate memory-like module. Recent work has demonstrated that organising internal representations into episodic structures and retrieving them selectively can substantially improve

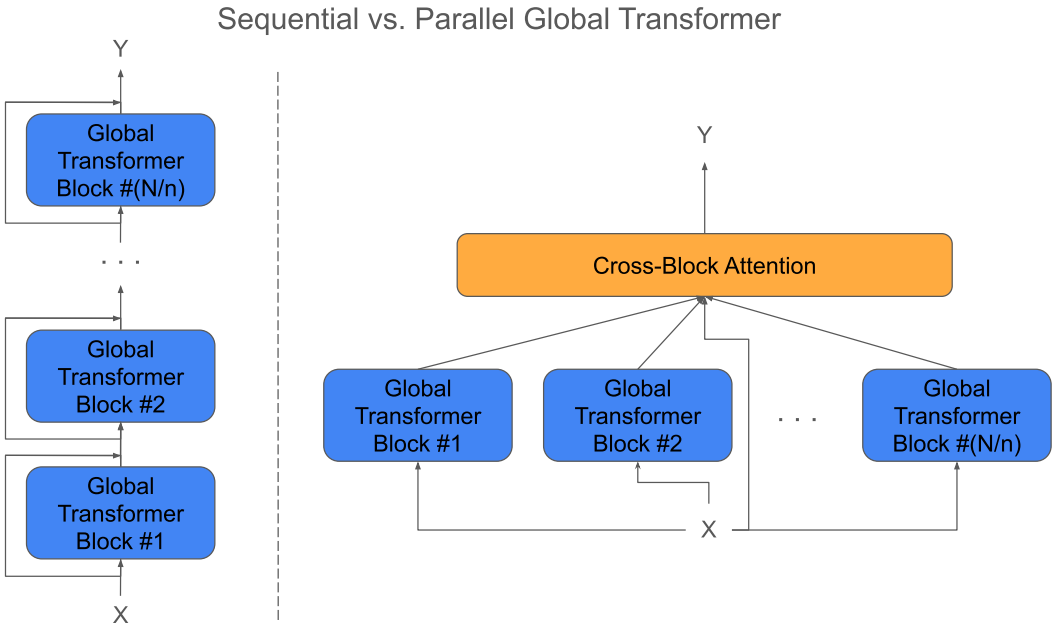


Figure 2: This figure illustrates two alternative formulations to the recurrent Global Transformer introduced in Fig. 1. (Left) The sequential formulation most closely resembles the vanilla Transformer architecture but is therefore also constrained to use a single residual stream between Global blocks. (Right) The parallel formulation allows much more sparse contributions to the output by each block but the fact that each block is fully independent from the others limits inter-block collaboration entirely to the shared workspace (the new Cross-Block Attention layer). Both architectures lose the bi-directional access characteristic across blocks which the recurrent version boasts, although they maintain it across time.

long-context performance [Fountas et al., 2024], while formal analyses of reflective memory dynamics suggest that iterative read-write interactions with such stores can support continual adaptation [Wang, 2025, Zhou et al., 2025]. Combining episodic retrieval with the periodic KV cache consolidation mechanisms explored by Oomerjee et al. [2025] could yield a persistent, continually updating workspace state that supports both in-context and continual learning.